International Journal of Mining Science and Technology 36 (2026) 957–971

Contents lists available at ScienceDirect

International Journal of Mining Science and Technology

journal homepage: www.elsevier.com/locate/ijmst

Strength and *GSI* of deep rock mass: A GZZ strength criterion- and non-associated flow rule-based constitutive model and its application

Xiangyang Wei [a], Wuqiang Cai [b,*], Hehua Zhu [a,c], Wenhao Liang [a,b], Xiaojun Wang [a], Jinfeng Xu [a], Fengshou Zhang [a]

[a] College of Civil Engineering, Tongji University, Shanghai 200092, China
[b] Shanghai Key Laboratory of Rail Infrastructure Durability and System Safety, Tongji University, Shanghai 201804, China
[c] State Key Laboratory for Disaster Reduction in Civil Engineering, Tongji University, Shanghai 200092, China

ARTICLE INFO

Article history:
Received 5 November 2025
Received in revised form 30 December 2025
Accepted 9 March 2026
Available online 25 March 2026

Keywords:
Deep rock mass
GZZ strength criterion
Equivalent fractured rock mass
True triaxial test
Digital in-situ testing

ABSTRACT

Accurate mechanical parameters are crucial for deep rock engineering. Traditional two-dimensional strength models often fail to reflect the complex three-dimensional mechanical properties of deep rock mass. This study proposes an elastoplastic constitutive model based on the smooth GZZ strength criterion, incorporating a non-associated flow rule to account for rock dilatancy. The model was numerically implemented and validated against theoretical and experimental results. Applied to a deep-buried tunnel, it determined the scale-dependent uniaxial compressive strength (UCS) and Geological Strength Index (*GSI*) of the rock mass. Numerical experiments revealed a transition from brittle to ductile failure with increasing confining pressure. Both the strength and *GSI* increase nonlinearly with confining pressure. The impacts of the plastic flow rule and rock matrix strength criterion on these parameters were quantitatively analyzed. At high confining pressures, neglecting dilatancy leads to their overestimation. Compared to the proposed model, using the Hoek-Brown criterion for the rock matrix fails to capture continuous hardening and yields conservative strength predictions under high confinement. Overall, the proposed model offers an improved tool for predicting rock mass strength and *GSI* under high confinement, with direct implications for the design and stability assessment of deep underground excavations.

© 2026 China University of Mining & Technology. Publishing services by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1. Introduction

The exploration of deep-earth science and engineering theories is critical for achieving the goal of "advancing into the deep Earth." Deep rock mass exhibits mechanical behavior markedly different from shallow rock mass [1–4], with nonlinear responses under high in-situ stress being particularly pronounced. However, accurately simulating the three-dimensional (3-D) and nonlinear mechanical behavior of surrounding rock under high-stress conditions remains a critical challenge due to its highly concealed nature and significant uncertainties.

Numerical simulation is a vital tool for rock engineering, relying on accurate mechanical models of rock mass [2,5]. For large-scale rock engineering, elastoplastic theories are commonly employed for mechanical analysis, where strength criteria serve as a core component for determining rock mass failure and assessing stability. To date, numerous strength criteria have been developed [6–10], however, each has limitations tied to lithology, stress level, and hydrogeological conditions. The most widely used Mohr–Coulomb (M–C) criterion, while owning clear physical parameters (cohesion c and internal friction angle φ), ignores the effect of intermediate principal stress, and cannot describe the nonlinear rock strength under high-stress conditions due to its linear assumption. The empirical Hoek–Brown (H–B) criterion, based on extensive triaxial rock tests and shallow rock data [11–13], introduces nonlinearity and a systematic parameter determination method but also neglects the intermediate principal stress. This limitation becomes more pronounced for deep rock mass under complex true 3-D stress states [1,14,15]. To address this, various 3-D criteria have been developed from 2-D ones [6,16]. Among them, the generalized Zhang–Zhu (GZZ) criterion accurately captures true triaxial test data while reducing to the H–B criterion under conventional triaxial conditions, thereby inheriting its reliable parameter determination method [16]. This is the primary reason for selecting the GZZ criterion in this study. In addition to

* Corresponding author.
E-mail address: caiwuqiang95@tongji.edu.cn (W. Cai).

https://doi.org/10.1016/j.ijmst.2026.03.006
2095-2686/© 2026 China University of Mining & Technology. Publishing services by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

static strength criteria, the development of dynamic models has also attracted attention [17,18]. Wang et al. [17] advanced the Holmquist–Johnson–Cook (HJC) model by integrating an additional failure criterion, which significantly improved its ability to characterize the tensile damage behavior of rocks, providing a key basis for developing more accurate dynamic models.

Reliable stability analysis requires not only accurate models but also rational rock mass parameters [12,19,20]. Generally, mechanical parameters of rock mass depend on both structural characteristics (internal factor) and stress level (external factor). In shallow engineering, low stress level means parameters are largely governed by rock mass structure and negatively correlated with the development degree of discontinuities [21]. While for deep rock mass, high stress reduces the weakening effect of discontinuities, manifested as: (1) weakened or vanished strength anisotropy [5], and (2) increased Geological Strength Index (*GSI*) with depth, indicating improved rock mass quality and integrity [15].

Common parameter estimation methods include in-situ experiments [22], empirical formulas [11,13], and numerical simulations. Among these, numerical methods are favored for their wide applicability, strong repeatability, and low costs. Recent studies combine equivalent rock mass techniques [23] with programs like UDEC [24], 3DEC [25], PFC [26], and RFPA [27] investigate size effect and anisotropy of rock mass, establishing a standard method for determining the Representative Elementary Volume (REV) and equivalent parameters. These studies predominantly rely on conventional 2-D strength criteria (e.g., M–C or H–B) for the rock matrix, which can yield relatively satisfactory results under low-stress, shallow conditions. However, as depth increases, the 3-D and nonlinear characteristics of rock become more pronounced. Therefore, there is a critical need to integrate advanced 3-D and nonlinear constitutive models for the rock matrix into the equivalent rock mass analysis framework, particularly for deep high-stress conditions. Furthermore, equivalent parameters from numerical studies often lack experimental validation. The Discrete Fracture Networks (DFN) generated using Monte-Carlo method always introduces uncertainty, as selected DFN models are statistical subsets that may not fully represent reality. Moreover, conducting simulations based on these DFN models would significantly increase computational costs [25]. To effectively control the errors arising from DFN models, Zhang et al. [28,29] proposed a dimension-raising method for geometric attributes of discontinuities that can estimate the real 3-D distribution of discontinuities based on low-dimensional data. However, this method has not yet been applied to estimate rock mass parameters.

In summary, accurate models and parameters are vital for stability assessment in deep underground engineering. While existing 3-D strength theories can describe the complex mechanical behavior of deep rock mass, methodologies for determining corresponding equivalent parameters remain underdeveloped. Previous numerical studies on equivalent rock mass predominantly rely on 2-D strength criteria, which are inadequate for accurately characterizing deep rock mass under high-stress conditions. To bridge this gap, this study proposes an elastoplastic constitutive model based on the smooth GZZ strength criterion, incorporating a non-associated flow rule to account for rock dilatancy. The model is numerically implemented (Section 2) and validated against experimental data (Section 3). Subsequently, it is applied to estimate the uniaxial compressive strength (UCS) and *GSI* of the rock mass in a deep-buried tunnel engineering (Section 4), which represents a primary focus and contribution of this work. Finally, the failure mode of the equivalent fractured rock mass is further analyzed, and the influences of different plastic flow rules and strength criteria adopted for the rock matrix on the predicted strength and *GSI* are investigated (Section 5).

2. Elastoplastic constitutive model and its numerical implementation

Taking into consideration the ductility characteristics of rocks under high in-situ stress environments and the robustness of numerical simulation, this paper adopts an ideal elastoplastic model to describe the stress–strain relationship of rock materials.

2.1. Smooth GZZ strength criterion

The Zhang–Zhu strength criterion, first proposed by Zhang and Zhu [16], is a 3-D nonlinear criterion for rock mass, which was later improved into the Generalized Zhang–Zhu (GZZ) strength criterion [30]. In order to further address the non-smooth and non-fully-convex issues of the GZZ strength criterion in numerical calculations, various modified criteria were proposed, including the smooth GZZ strength criterion [31]. Specifically, the original GZZ criterion is non-smooth (lacks continuous differentiability) at Lode angles of ±30°, which can cause convergence issues in stress integration, and is non-fully-convex near 30°, potentially violating the Drucker postulate. The smooth version overcomes these limitations by ensuring a continuously differentiable and fully convex yield surface, thereby providing a robust foundation for the present elastoplastic implementation. The smooth GZZ criterion is expressed as

$$F = \sigma_{ci}\left(\frac{\sqrt{3J_2}}{\sigma_{ci}}\right)^{1/a} + \left(\frac{3+\sin 3\theta_\sigma}{2\sqrt{3}}\right)m_b\sqrt{J_2} - m_b\frac{I_1}{3} - s\sigma_{ci} \tag{1}$$

where the invariants $I_1 = \sigma_1+\sigma_2+\sigma_3$; $J_2 = \frac{1}{6}\left[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2\right]$; $\theta_\sigma = \tan^{-1}\left[\frac{2\sigma_2-(\sigma_1+\sigma_3)}{\sqrt{3}(\sigma_1-\sigma_3)}\right]$ signifies Lode angle; σ_{ci} the UCS of intact rock; and m_b, s and a the strength parameters of the H–B strength criterion.

It is worth noting that when Lode angle θ_σ is taken as ± 30° (i.e., triaxial tension or compression), the GZZ strength criterion is completely degenerated into the H–B strength criterion, the 3-D form of which is

$$F = \sigma_{ci}\left(\frac{2\cos\theta_\sigma\sqrt{J_2}}{\sigma_{ci}}\right)^{1/a} + \left(\cos\theta_\sigma + \frac{\sin\theta_\sigma}{\sqrt{3}}\right)m_b\sqrt{J_2} - m_b\frac{I_1}{3} - s\sigma_{ci} \tag{2}$$

Therefore, the GZZ strength criterion adopts the parameter determination method inherited from the existing H–B strength criterion, that is

$$\begin{cases} m_b = m_i \exp\left(\frac{GSI-100}{28-14D}\right) \\ s = \exp\left(\frac{GSI-100}{9-3D}\right) \\ a = \frac{1}{2}+\frac{1}{6}\left[\exp\left(-\frac{GSI}{15}\right)-\exp\left(-\frac{20}{3}\right)\right] \end{cases} \tag{3}$$

where m_i signifies the material parameter that characterizes the quality of intact rock, which can be obtained through triaxial compression tests or selected by referring to the table given by Hoek [11]; D the engineering disturbance coefficient; and *GSI* the geological strength index.

2.2. Non-associated flow rule

Plastic flow rule is an important component of elastoplastic constitutive models. Due to the fact that the rock mass deformation calculated with associated flow rule tends to be overestimated, a non-associated flow rule is usually adopted to improve the calculation accuracy [32,33]. Currently, the widely accepted non-associated flow rules are primarily established based on strength criteria, such as the M–C criterion, the H–B criterion, and the GZZ criterion. Deep rock mass generally exhibits signifi-

cant plastic volumetric deformation under excavation; therefore, the elastoplastic constitutive model should also employ a non-associated flow rule. However, there is currently a lack of a non-associated flow rule compatible with the smooth GZZ strength criterion, along with its numerical implementation, which constitutes the fundamental issue to be addressed in this paper.

2.2.1. Plastic potential function

In the GZZ strength criterion, the plastic volumetric deformation is primarily governed by $F_1(I_1) = -m_b \frac{I_1}{3}$ [33]. By introducing a correction coefficient into this term that reflects rock dilatancy effects, the plastic potential function combining a non-associated flow rule can be obtained as

$$Q = \sigma_{ci}\left(\frac{\sqrt{3J_2}}{\sigma_{ci}}\right)^{1/a} + \left(\frac{3+\sin 3\theta_\sigma}{2\sqrt{3}}\right)m_b\sqrt{J_2} - \beta m_b\frac{I_1}{3} - s\sigma_{ci} \tag{4}$$

where β is the non-associated flow coefficient, which governs the plastic volumetric behavior by defining the deviation between the plastic potential and yield surfaces. Its value is typically calibrated against laboratory triaxial test data or through numerical back-analysis of field observations.

2.2.2. Plastic flow vector

The relation between Lode angle θ_σ and the invariants of deviatoric stress tensor, J_2 and J_3, is

$$\sin 3\theta_\sigma = -\frac{3\sqrt{3}}{2}\cdot\frac{J_3}{J_2^{3/2}} \tag{5}$$

Therefore, Eq. (4) can be rearranged as

$$Q = \frac{\left(\sqrt{3J_2}\right)^{1/a}}{\sigma_{ci}^{1/a-1}} + \left(\frac{\sqrt{3J_2}}{2} - \frac{3J_3}{4J_2}\right)m_b - \beta m_b\frac{I_1}{3} - s\sigma_{ci} \tag{6}$$

The expression for plastic flow vector can be written as

$$\frac{\partial Q}{\partial \boldsymbol{\sigma}} = \frac{\partial Q}{\partial I_1}\frac{\partial I_1}{\partial \boldsymbol{\sigma}} + \frac{\partial Q}{\partial \sqrt{J_2}}\frac{\partial \sqrt{J_2}}{\partial \boldsymbol{\sigma}} + \frac{\partial Q}{\partial J_3}\frac{\partial J_3}{\partial \boldsymbol{\sigma}} \tag{7}$$

According to Eq. (6), the partial derivatives of the plastic potential function with respect to each stress invariant are determined by

$$\begin{cases} \frac{\partial Q}{\partial I_1} = -\frac{\beta}{3}m_b \\ \frac{\partial Q}{\partial \sqrt{J_2}} = h(J_2,\theta_\sigma)m_b \\ \frac{\partial Q}{\partial J_3} = -\frac{3}{4J_2}m_b \end{cases} \tag{8}$$

and the partial derivatives of each stress invariant can be derived as

$$\begin{cases} \frac{\partial I_1}{\partial \boldsymbol{\sigma}} = \boldsymbol{I} \\ \frac{\partial \sqrt{J_2}}{\partial \boldsymbol{\sigma}} = \frac{\boldsymbol{S}}{2\sqrt{J_2}} \\ \frac{\partial J_3}{\partial \boldsymbol{\sigma}} = \boldsymbol{S}\cdot\boldsymbol{S} - \frac{2J_2}{3}\boldsymbol{I} \end{cases} \tag{9}$$

where $h(J_2,\theta_\sigma) = \sqrt{3}\left[\frac{1}{am_b}\left(\frac{\sqrt{3J_2}}{\sigma_{ci}}\right)^{1/a-1} - \frac{\sin 3\theta_\sigma}{3} + \frac{1}{2}\right]$; and $\boldsymbol{I}$ and $\boldsymbol{S}$ the unit tensor and deviatoric stress tensor, respectively.

By substituting Eqs. (8) and (9) into Eq. (7) and simplifying them, the plastic flow vector is obtained as

$$\frac{\partial Q}{\partial \boldsymbol{\sigma}} = m_b\left[\left(\frac{1}{2}-\frac{\beta}{3}\right)\boldsymbol{I} + \frac{h(J_2,\theta_\sigma)}{2\sqrt{J_2}}\boldsymbol{S} - \frac{3}{4J_2}\boldsymbol{S}\cdot\boldsymbol{S}\right] \tag{10}$$

2.2.3. Plastic strain increment and plastic volumetric strain increment

In the light of the principles of geotechnical plasticity, the relation between the plastic strain increment and the plastic flow vector is derived as

$$\mathrm{d}\boldsymbol{\varepsilon}^p = \lambda\frac{\partial Q}{\partial \boldsymbol{\sigma}} = \lambda m_b\left[\left(\frac{1}{2}-\frac{\beta}{3}\right)\boldsymbol{I} + \frac{h(J_2,\theta_\sigma)}{2\sqrt{J_2}}\boldsymbol{S} - \frac{3}{4J_2}\boldsymbol{S}\cdot\boldsymbol{S}\right] \tag{11}$$

where λ denotes plastic multiplier.

Furthermore, the plastic volumetric strain increment is expressed as

$$\mathrm{d}\varepsilon_v^p = \mathrm{d}\varepsilon_{ii}^p = -\lambda\beta m_b \tag{12}$$

Eq. (12) shows that the plastic volumetric strain increment is proportional to the non-associated flow coefficient β. When $\beta = 0$, $\mathrm{d}\varepsilon_v^p = 0$, indicating a constant-volume flow rule; while when $\beta = 1$, associated flow rule is followed, i.e., $Q = F$.

2.3. Numerical implementation

The aforementioned constitutive model is implemented via the User-Defined Constitutive Model (UDM) feature within the 3-D discrete element code 3DEC 7.0 [34], which utilizes an explicit finite-difference solution scheme. In this section, four key steps are presented.

Step 1: Calculation of trial stresses

$$\boldsymbol{\sigma}^{\mathrm{trial}} = \boldsymbol{\sigma}^n + \mathbb{C} : \Delta\boldsymbol{\varepsilon} \tag{13}$$

where the fourth-order tensor $\mathbb{C}$ is elastic tensor; $\boldsymbol{\sigma}^{\mathrm{trial}}$ the trial stress tensor; $\boldsymbol{\sigma}^n$ the stress tensor from the last time-step; and $\Delta\boldsymbol{\varepsilon}$ the strain increment tensor.

Step 2: Plastic correction judgment.

The yield function value at the trial stresses, F^{trial}, is calculated first according to Eq. (1). If $F^{\mathrm{trial}} < 0$, no correction will be made, and the current stresses are equal to the trial stresses. Otherwise, Step (3) is required.

Step 3: Determination of plastic multiplier λ.

The stress integration follows an explicit (forward Euler) scheme. The stability of this explicit update is ensured by the extremely small, automatically determined critical time-step used in 3DEC's dynamic solution. Within this framework, the plastic multiplier λ can be efficiently approximated using the first-order Taylor expansion at the trial stress state.

The expression for stress correction should be declared first, that is

$$\boldsymbol{\sigma}^{n+1} = \boldsymbol{\sigma}^{\mathrm{trial}} - \lambda\mathbb{C} : \boldsymbol{n}_g \tag{14}$$

where $\boldsymbol{\sigma}^{n+1}$ signifies the stress tensor after correction; and $\boldsymbol{n}_g = \frac{\mathrm{d}Q}{\mathrm{d}\boldsymbol{\sigma}}\big|_{\boldsymbol{\sigma}^{\mathrm{trial}}}$ the gradient of plastic potential function at the trial stresses, which is determined using Eq. (10).

The derivative of the yield function with respect to λ is estimated by

$$\frac{\mathrm{d}F}{\mathrm{d}\lambda} = \frac{\mathrm{d}F}{\mathrm{d}\boldsymbol{\sigma}} : \frac{\mathrm{d}\boldsymbol{\sigma}}{\mathrm{d}\lambda} \tag{15}$$

Given the small time-step, λ is approximated using the first-order derivative of the yield function at the trial stresses. The expression is given as

$$\lambda = -F^{\mathrm{trial}} \Big/ \left(\frac{\mathrm{d}F}{\mathrm{d}\lambda}\Big|_{\boldsymbol{\sigma}^{\mathrm{trial}}}\right) = \frac{F^{\mathrm{trial}}}{\boldsymbol{n}_f : \mathbb{C} : \boldsymbol{n}_g} \tag{16}$$

where $\boldsymbol{n}_f = \frac{\mathrm{d}F}{\mathrm{d}\boldsymbol{\sigma}}\big|_{\boldsymbol{\sigma}^{\mathrm{trial}}}$ is the gradient of yield function at the trial stresses, which is determined by Eq. (10) (Note that $\beta = 1$).

Step 4: Stress apex correction

The corrected stresses can be estimated by substituting λ obtained from Eq. (16) into Eq. (14). Under the condition that compression is positive, if $\sigma_1^{n+1} = \sigma_2^{n+1} = \sigma_3^{n+1} \leqslant -\frac{s\sigma_{ci}}{m_b}$ is satisfied, vertex correction should be performed. The stresses after apex correction are

$$\sigma_1^{n+1} = \sigma_2^{n+1} = \sigma_3^{n+1} = -\frac{s\sigma_{ci}}{m_b} \tag{17}$$

Note that since the GZZ strength criterion itself is a closed surface near the apex, and surrounding rock is unlikely to undergo tensile failure under deep high-stress environments, there is no need to set a tensile cutoff.

3. Laboratory tests and model validation

The following model validation and field application are based on a deep-buried railway tunnel project in western China. The design tunnel diameter is around 9 m and the maximum burial depth exceeds 2,000 m. According to in-situ stress measurement results, the maximum horizontal principal stress in this area ranges from 11.93 to 43.46 MPa. Additionally, the rocks are primarily hard rocks such as gneiss and granite, where the problem of rock burst is more prominent under high geo-stress conditions.

3.1. Laboratory experiments on intact rock and prefabricated fractured rock specimens

Homogeneous granite samples collected from the tunnel field were processed into standard intact rock specimens which measure 50 mm × 50 mm × 100 mm, along with two types of rock specimens containing prefabricated X-shaped fractures. The X-shaped fracture pattern was adopted as it represents a typical intersecting configuration of two dominant discontinuity sets commonly observed in rock mass. This controlled geometry serves as a fundamental and reproducible physical analogue for validating the subsequent numerical models, while capturing the essential mechanical interactions induced by cross-cutting fractures. As illustrated in Fig. 1, the rock specimens in case-1 include one set of prefabricated X-shaped fractures, while those in case-2 contain four sets of prefabricated X-shaped fractures.

For the intact rock specimens, uniaxial compression tests were conducted to determine the UCS, the elastic modulus, and the Poisson's ratio. Subsequently, two sets of true triaxial compression tests ($\sigma_2 = 3\sigma_3 = 0.3\sigma_{ci}$ and $\sigma_2 = 5\sigma_3 = 0.5\sigma_{ci}$) were carried out to obtain the true triaxial strength. Furthermore, m_i was solved based on Eqs. (1) and (3) for both tests, and its average was taken as the final value. Note that for intact rock, $GSI = 100, D = 0$. Table 1 lists the mechanical parameters of intact rock.

Subsequently, uniaxial and true triaxial compression experiments under two stress paths ($\sigma_2 = 3\sigma_3 = 0.3\sigma_c$ and $\sigma_2 = 5\sigma_3 = 0.5\sigma_c$) were conducted on the rock specimens in case-1 and case-2, respectively, where σ_c signifies the UCS of the fractured specimen in each case. The corresponding strengths of the rock specimens with different sets of prefabricated X-shaped fractures were recorded and presented in Table 2.

3.2. Model validation with intact rock

Numerical experiments involving uniaxial and conventional triaxial compression (with confining pressures of 20, 40, and 60 MPa) were conducted on the intact rock. The confining pressure was applied via surface stress loading, while axial loading was implemented under displacement control at a rate of 0.01 m/s. The resulting stress–strain curves are plotted in Fig. 2, and the peak values were extracted as the strength. Comparisons show that the numerical strength values of intact rock under different confining pressures align well with the theoretical results calculated directly using the GZZ strength criterion (with $GSI = 100$).

In addition, the volumetric deformation curves of intact rock under different confining pressures, when using constant-volume law and associated law respectively, are presented in Fig. 3a–b. The results indicate that under constant-volume flow rule, the volumetric deformation of intact rock remains constant after the peak strength, making it suitable for high confining pressure conditions. In contrast, associated flow rule can capture the post-peak dilatancy behavior of intact rock, rendering it more applicable to low confining pressure scenarios.

Furthermore, for the rock using the GZZ strength criterion and the H–B strength criterion respectively, numerical true triaxial tests were conducted under different Lode angles (0°, ±10°, ±20°, and ± 30°) at a stress level of $I_1 = 360$ MPa. The strength $\sqrt{J_2}$ on the deviatoric plane was obtained, as displayed in Fig. 4. It can be observed that all the numerical strength points under different strength criteria fall on their corresponding theoretical strength curves, which confirms the successful numerical implementation of the constitutive models. Moreover, at the Lode angle of ±30°,

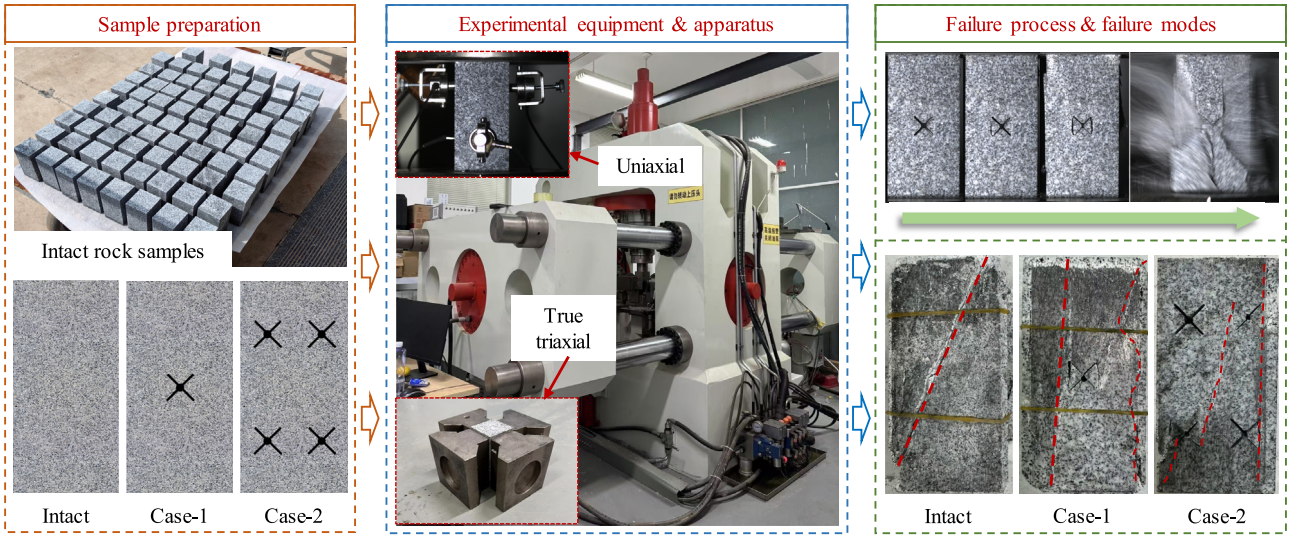


Fig. 1. Uniaxial and true triaxial compression experiments on the rock specimens.

Table 1
Mechanical properties of intact rock.

Property	Value
Density (kg/m^3)	2.7×10^3
Elastic modulus, E (GPa)	66.2
Poisson's ratio, ν	0.22
UCS, σ_{ci} (MPa)	168.4
Material constant,m_i	23.1
Geological strength index, *GSI*	100
Disturbance factor, D	0
Non-associated flow coefficient, β	0

Table 2
Experimental strength of rock specimens in different cases.

Loading condition	Strength (MPa)		
	Intact rock	Case-1 (one set of X-shaped fractures)	Case-2 (four sets of X-shaped fractures)
$\sigma_2 = \sigma_3 = 0$	168.4	131.1	65.5
$\sigma_2 = 3\sigma_3 = 0.3\sigma_c$	536.0	316.1	173.7
$\sigma_2 = 5\sigma_3 = 0.5\sigma_c$	622.1	340.6	153.1

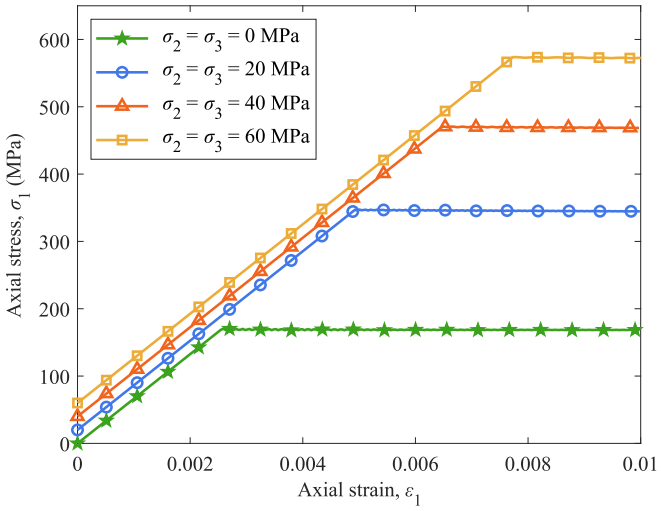


Fig. 2. Axial stress–strain curves of the intact rock under compression.

the strengths obtained using both strength criterion are equal, indicating that the GZZ criterion can fully reduce to the H–B strength criterion under triaxial tension or compression conditions.

3.3. Model validation with prefabricated fractured rock specimens

Equivalent numerical models of rock mass were established for both case-1 and case-2. For comparison, the GZZ strength criterion and the H–B strength criterion were adopted for the rock matrix, respectively, with mechanical parameters listed in Table 1. The fractures were simulated using the M–C model, wherein the normal stiffness and tangential stiffness of virtual fractures were set to 100 times the elastic modulus and shear modulus of rock matrix, respectively. The strength parameters were approximated by the tangent of the H–B criterion at $\sigma_3 = 0$, and the tensile strength was taken as the strength at the apex, that is

$$c_{\text{equiv}} = \frac{\sigma_{ci}}{\sqrt{2(2+m_i)}}, \varphi_{\text{equiv}} = \sin^{-1}\frac{m_i}{m_i+4}, \sigma_{t,\text{equiv}} = \frac{\sigma_{ci}}{m_i} \quad (18)$$

For prefabricated fractures, to prevent mutual embedding of adjacent blocks in numerical simulations, the normal stiffness was set to 5 times the elastic modulus of rock, while the tangential stiffness was taken as 0.1 times the normal stiffness. Additionally, due to the large aperture of the prefabricated fractures (approximately 2 mm), the strength parameters were set as $c = \varphi = \sigma_t = 0$. The relevant mechanical parameters for both virtual and prefabricated fractures are provided in Table 3.

Uniaxial and true triaxial compression experiments following the same stress paths as laboratory tests were further conducted for the two numerical models (case-1 and case-2), respectively, and the corresponding strength properties of the numerical rock specimens were obtained. Fig. 5a–b presents a comparison between the numerical and experimental results for case-1 and case-2, respectively. Intuitively, the predicted strength of the rock specimens using the GZZ strength criterion for the rock matrix is closer to the experimental results than that using the H–B strength criterion, particularly under true triaxial loading conditions.

Fig. 6a–b illustrates the prediction errors of strength in case-1 and case-2 relative to the experimental values, respectively. For case-1, as shown in Fig. 6a, the prediction errors when the GZZ criterion is used for the rock matrix are all smaller than those when using the H–B criterion under three loading conditions. Specifically, the maximum prediction error when employing the GZZ criterion is 10.2%. In contrast, the minimum prediction error when using the H–B criterion is 33.6%. Additionally, compared to the uniaxial compression results, the prediction error when using the GZZ criterion under true triaxial compression conditions is smaller, while the error when using the H–B criterion is larger. For case-2, under uniaxial compression conditions, the prediction error when using the GZZ criterion is 22.0%, which is higher than the error of 11.6% when using the H–B criterion, as indicated in Fig. 6b. It may be attributed to experimental errors, and the relatively small absolute value of the experimental strength of rock specimen (65.5 MPa) can also be one of the reasons for the high relative error. Furthermore, under true triaxial compression conditions, similar to the results obtained for case-1, the prediction error when utilizing the GZZ criterion for the rock matrix is significantly lower than that when using the H–B criterion. This is because the GZZ strength criterion takes into account the positive influence of intermediate principal stress.

From above, the proposed elastoplastic constitutive model based on the smooth GZZ strength criterion can effectively capture the strength and dilatancy characteristics of intact rock. Furthermore, compared to the H–B strength criterion, using the GZZ strength criterion for the rock matrix can yield predictions of strength that align more closely with the experimental results of prefabricated fractured rock specimens. Therefore, the constitutive model proposed in this study can be applied to estimate the strength and *GSI* of fractured rock mass.

4. UCS and *GSI* of deep rock mass

Following the validation presented above, this section employs the proposed model directly for analyzing the equivalent mechanical parameters of surrounding rock in deep-buried tunnels.

4.1. Numerical modeling of equivalent fractured rock mass

Aiming at the aforementioned deep-buried tunnel project, Fig. 7 illustrates the schematic diagram of the digital-numerical integrated modeling process for equivalent fractured rock mass. The procedure consists of three key steps:

(1) Rock mass information acquisition and identification: multi-angle digital images of consecutive tunnel excavation faces were

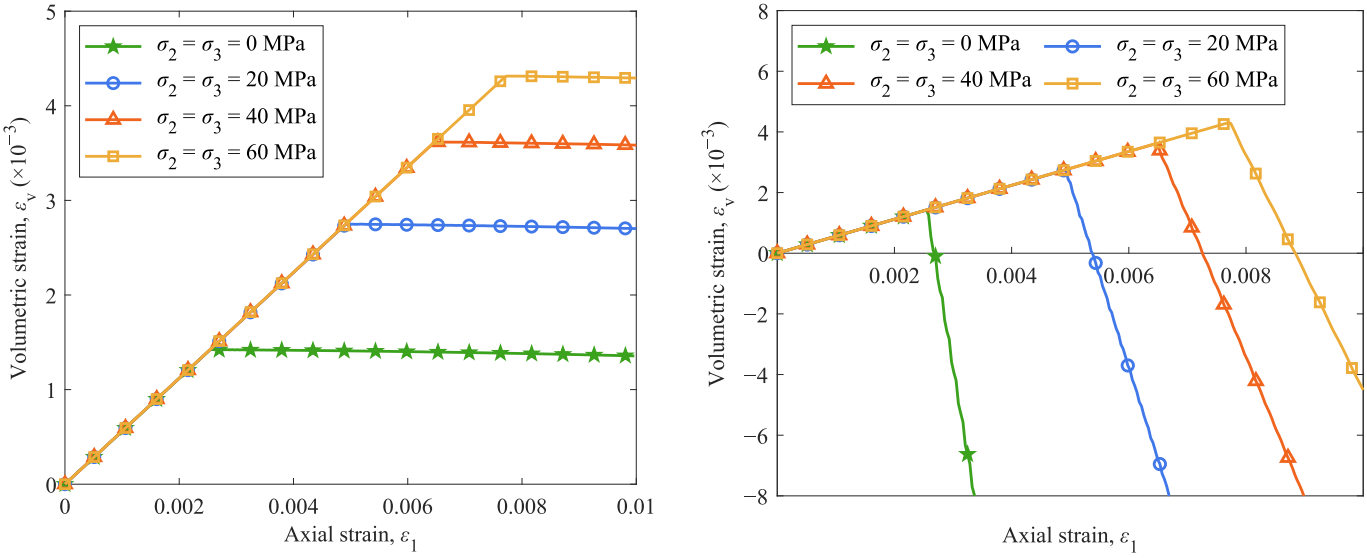


Fig. 3. Volumetric deformation curves of the intact rock under compression.

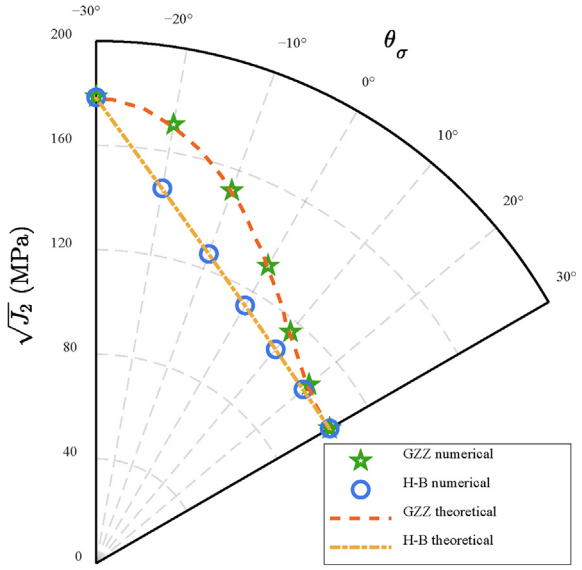


Fig. 4. Rock strength on the deviatoric plane ($I_1 = 360$ MPa).

Table 3
Mechanical properties of fractures.

Property	Fictitious fractures	Prefabricated fractures	Actual fractures
Normal stiffness, k_n (GPa/m)	6.62×10^3	331	331
Shear stiffness, k_s (GPa/m)	2.71×10^3	33.1	33.1
Cohesion, c (MPa)	23.77	0	
Friction angle, φ (°)	58.47	0	
Tensile strength, σ_t (MPa)	7.29	0	0
Joint roughness coefficient, *JRC*			3.78
Joint compressive strength, *JCS* (MPa)			31.8
Residual friction angle, φ_{res} (°)			18.1

collected with a smart phone. 3-D point cloud data for each excavation face was then obtained using an open-source computer vision platform, Meshroom, and discontinuities identification algorithms (including K-means, CFSFDP, and spectral clustering) were employed to extract geometric parameters including orientation, trace length, and spacing of discontinuities [35].

(2) 3-D DFN modeling: based on the 2-D statistical data from the excavation face, dimension-raising technology for geometric attributes of discontinuities (using Bayesian law) [28] was applied to estimate the 3-D hybrid distribution of geometric parameters, establishing a DFN model that can accurately characterize the actual distribution of discontinuities in the rock mass ahead of the excavation face. The effective modeling domain measures 6 m × 12 m × 10 m. This advanced technology enables unbiased and rapid estimation of geometric parameters of discontinuities in 3-D space, effectively mitigating computational errors caused by inaccuracies in DFN modeling.

(3) Equivalent fractured rock mass sampling: To capture the size effect and spatial heterogeneity, five sampling points were symmetrically arranged within the DFN model domain. Using the closest excavation face as the rock mass boundary and advancing along the excavation direction, a total of 30 equivalent fractured rock mass samples with the size ranging from 1 m to 6 m were obtained through cutting operations.

4.2. Model parameters

The parameters used for the rock matrix mainly come from laboratory rock mechanics tests, as provided in Table 1. As for fractures, the parameter determination method for virtual fractures has been introduced in Section 3.3, and will not be repeated here. For real fractures, considering the convenience of parameter acquisition, the JRC–JCS strength criterion is adopted in this paper [36], which is given by

$$\tau = \sigma_n \tan\left[JRC \cdot \log_{10}\left(\frac{JCS}{\sigma_n}\right) + \varphi_{res}\right] \tag{19}$$

The parameters of the JRC–JCS model include joint roughness coefficient (*JRC*), joint compressive strength (*JCS*), and residual friction angle (φ_{res}). A digital method is employed herein to estimate the *JRC* of discontinuities. As illustrated in Fig. 8, multi-angle images of numerous discontinuities were captured through photographic documentation on-site. 3-D point clouds were subsequently extracted from these images, and the average *JRC* was calculated using the formula in ref [37]. The *JCS* was derived through interpolation between the *JRC* and the UCS of intact rock, specifically

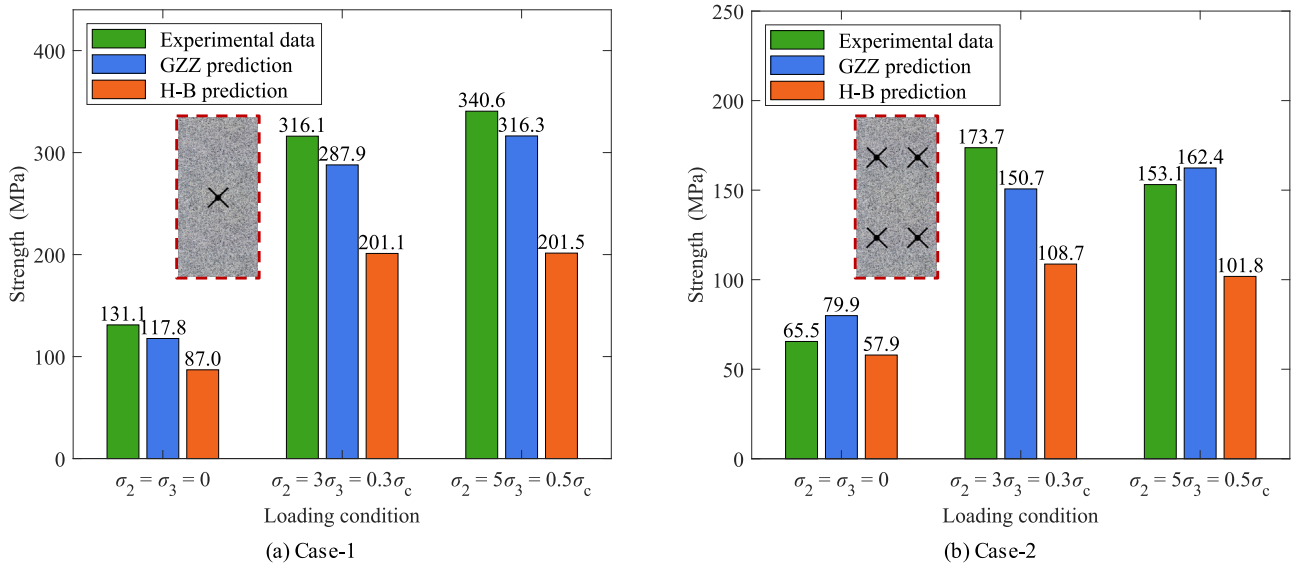


Fig. 5. Strength of the prefabricated fractured rock specimens.

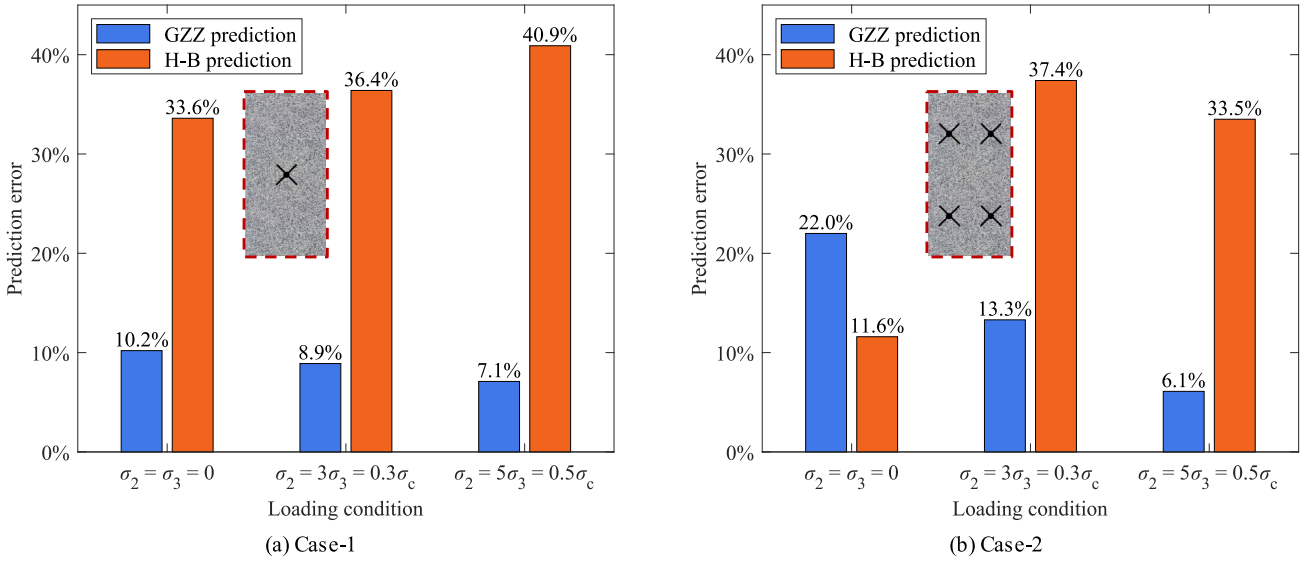


Fig. 6. Prediction error of strength of the prefabricated fractured rock specimens.

$$JCS = \frac{JRC}{20}\sigma_{ci} \tag{20}$$

The residual friction angle of discontinuities is determined by

$$\varphi_{res} = \left(\varphi_b - 20^\circ\right) + 20^\circ \times \frac{\log_{10}(JCS) - 1.01}{\log_{10}(\sigma_{ci}) - 1.01} \tag{21}$$

where the basic friction angle of intact rock is denoted as φ_b, and according to the table [38], φ_b of fine-grained granite is approximately 30°. Finally, the main mechanical parameters of virtual and real discontinuities are summarized in Table 3.

4.3. Numerical results

For the 30 equivalent fractured rock mass models, uniaxial compression experiments were conducted numerically along the X, Y, and Z directions using the model parameters listed in Tables 1 and 2. The average axial stress was calculated through dividing the support reactions at the bottom by the cross-sectional area, and the stress–strain curve for each equivalent fractured rock mass was recorded, from which the UCS was extracted. Furthermore, the parameter *GSI* was solved using the GZZ strength criterion. Finally, the relation between the UCS of rock mass and its size is plotted in Fig. 9a, and that between *GSI* and size is displayed in Fig. 9b.

As can be observed from Fig. 9, both the UCS and *GSI* fluctuations of rock mass show a decrease trend with increasing rock mass size. When the size exceeds a specific threshold, the two parameters stabilize with negligible variation, demonstrating significant size effect. Table 4 presents the maximum, mean, median, and minimum values of the UCS and *GSI* for different rock mass sizes. Analysis of the mean and median values reveals that for the size ranging from 3 m to 6 m, the mean UCS ranges from 39.1 to 40.2 MPa, the median UCS ranges from 39.6 to 41.4 MPa, while the mean *GSI* ranges from 73.7 to 74.0, and the median *GSI* ranges from 74.0 to 74.8. All the fluctuations are quite narrow.

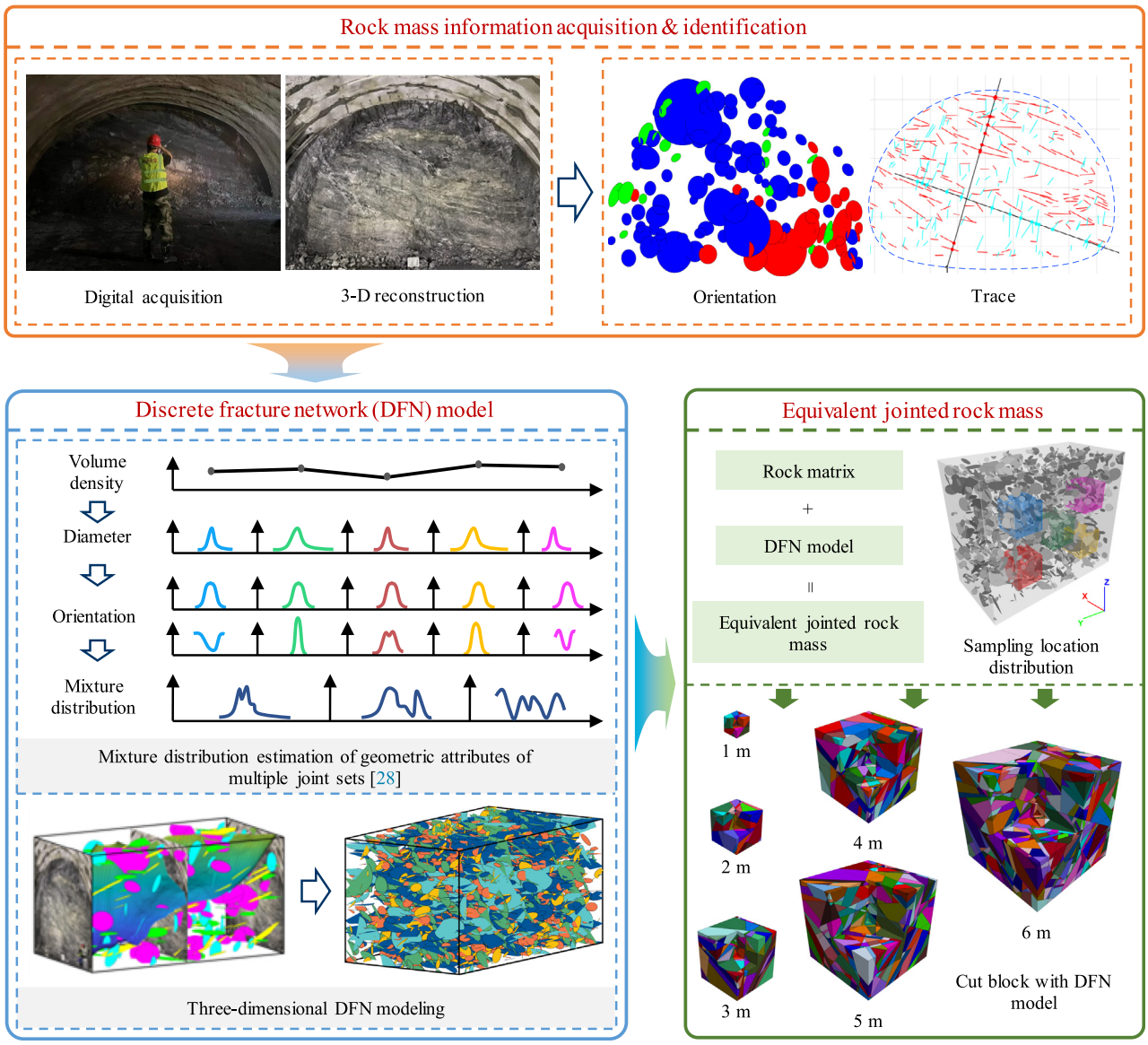


Fig. 7. Schematic diagram of numerical modeling of equivalent fractured rock mass.

From the perspective of maximum and minimum values, when the rock mass size reaches 5 m and 6 m, the fluctuation range of the predicted UCS are 31.1–45.7 MPa and 33.3–44.6 MPa, respectively. The corresponding fluctuation range of *GSI* are 69.7–75.5 and 70.9–76.1, respectively. This indicates that the upper and lower limits of both the UCS and *GSI* also tend to stabilize as the rock mass size increases.

In summary, the UCS and *GSI* of rock mass can be successfully determined by applying the proposed GZZ strength criterion-based model to the rock matrix. For future rock mass classification or further excavation simulation, the average UCS of 39.1 MPa and the average *GSI* of 73.7 for the equivalent fractured rock mass with a dimension of 6 m can be selected as design parameters. Compared to the strength and *GSI* of rock matrix, these values decreased by 76.8% and 26.3%, respectively. Alternatively, conservatively speaking, the lower limits of the UCS (33.3 MPa) and *GSI* (70.9) can be adopted as design parameters, corresponding to reductions of 80.2% and 29.1%, respectively.

5. Results and analysis: Failure mode and parameter effects

The above findings indicate that the equivalent fractured rock mass with a dimension of 3 m has essentially reached the REV level. Therefore, based on considerations of representativeness, computational cost and article length limitations, a representative equivalent fractured rock mass with a dimension of 3 m was selected for further investigation in this section. Numerical experiments under uniaxial and conventional triaxial compression (with confining pressures ranging from 5 to 60 MPa) were carried out.

5.1. Failure mode of rock mass

Figs. 10 and 11 illustrate the failure states of the rock matrix and the discontinuities (using constant-volume flow rule for the rock matrix), respectively. For the discontinuities, the failure is categorized into two states: tensile failure and shear failure, as shown in Fig. 10. While for the rock matrix, based on its current stress Lode angle, the failure modes are classified into six states, as

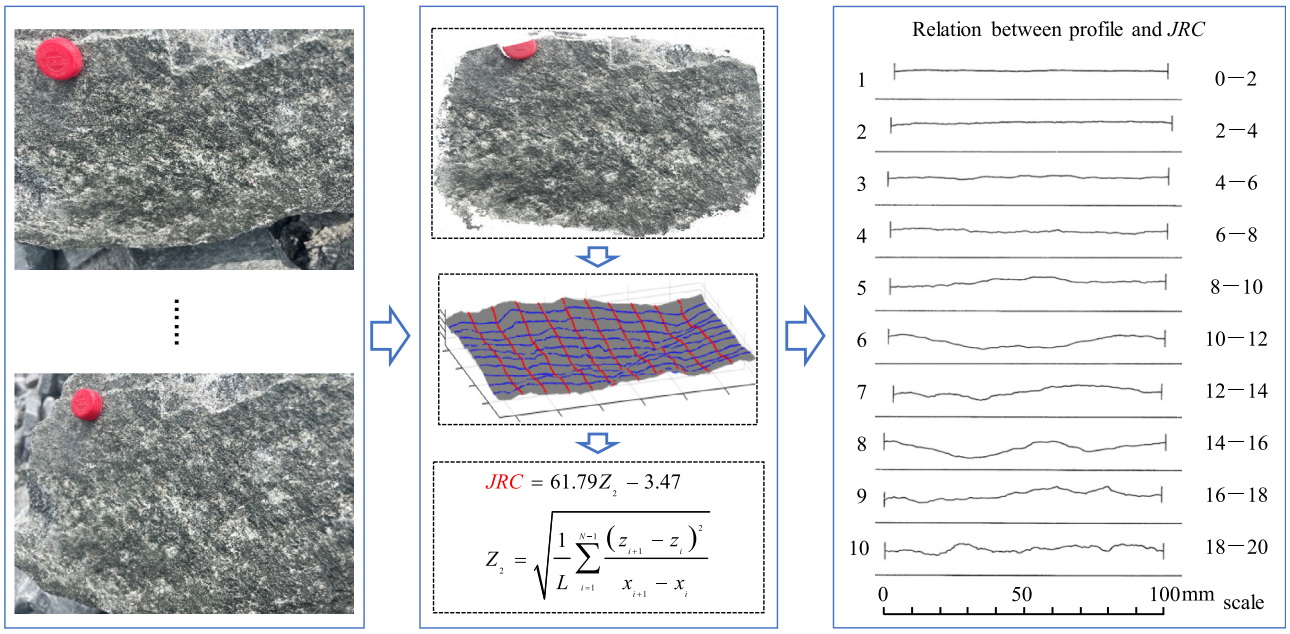


Fig. 8. Schematic diagram of the digital method for joint roughness coefficient (*JRC*) determination.

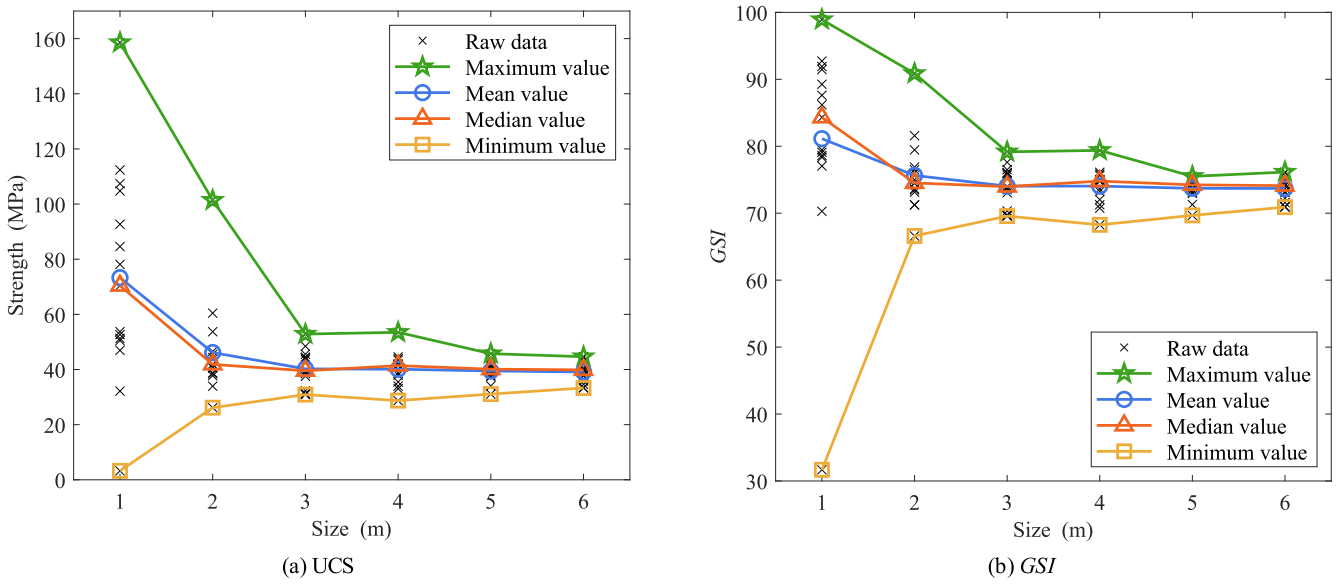
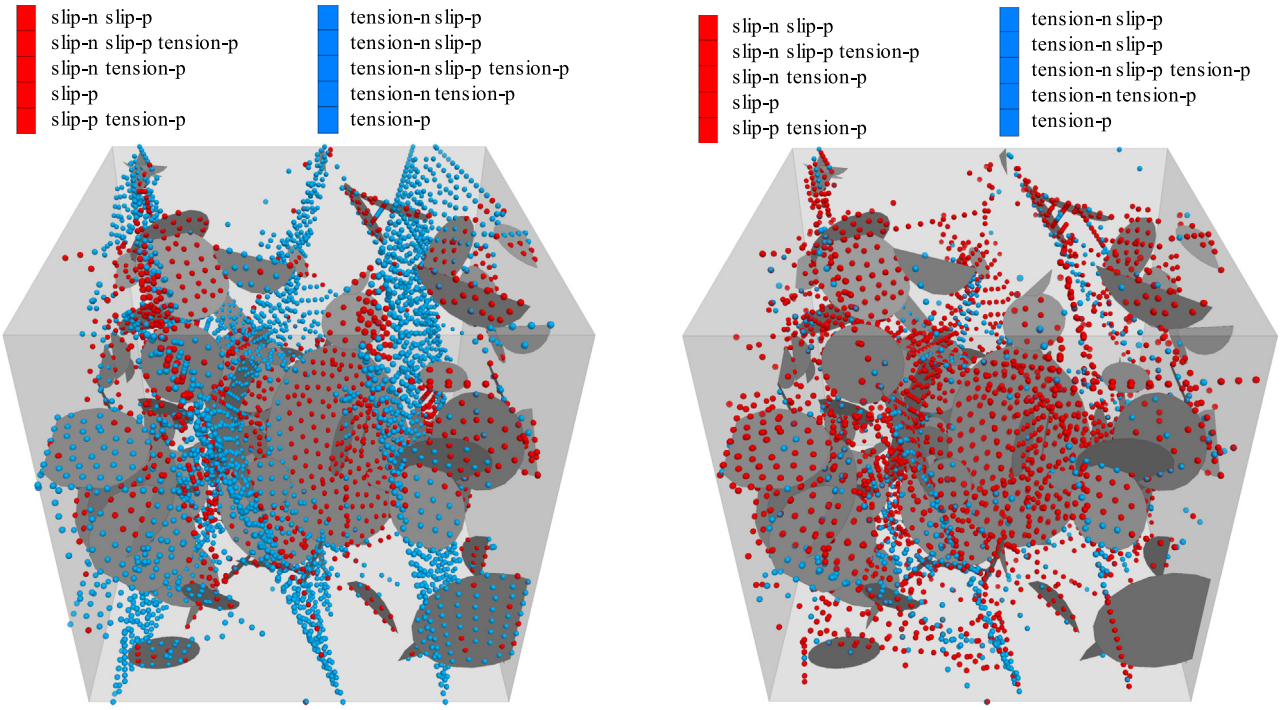


Fig. 9. Variation of equivalent rock mass parameters with rock mass size.

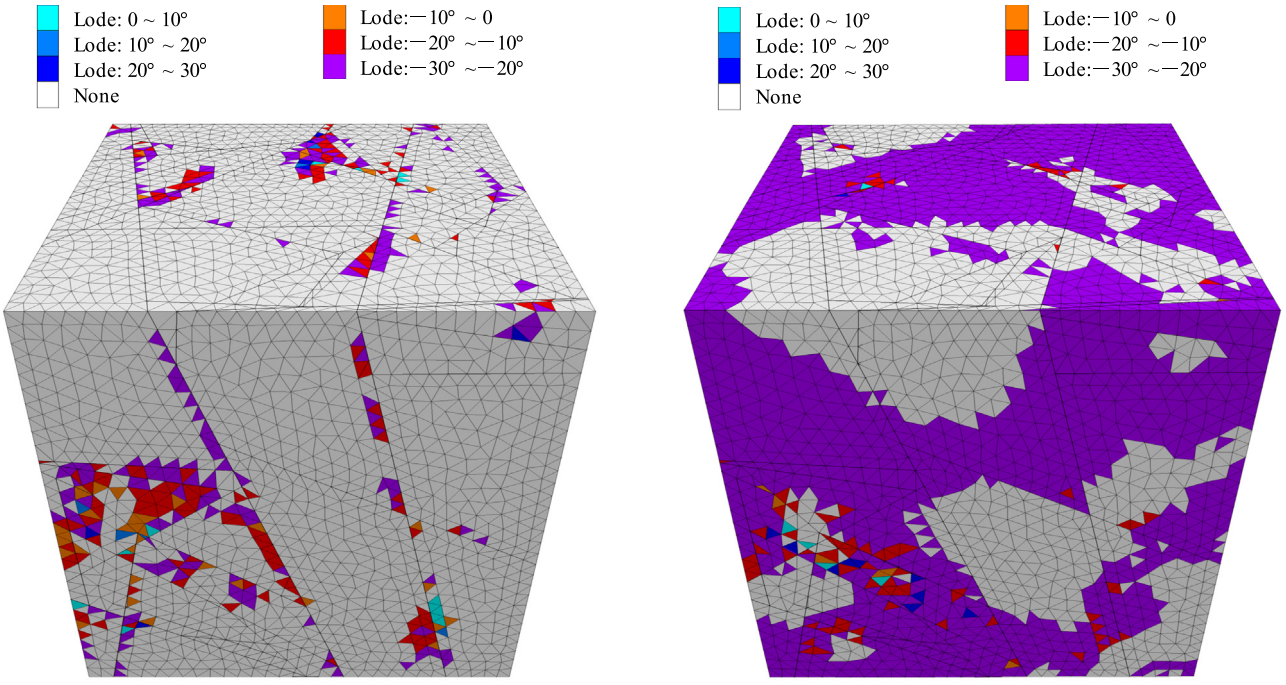
Table 4
UCS and *GSI* of rock mass with various rock mass sizes.

Size (m)	UCS (MPa)				*GSI*			
	Maximum	Average	Median	Minimum	Maximum	Average	Median	Minimum
1	158.6	73.3	70.4	3.3	98.9	81.1	84.3	31.7
2	101.4	46.2	41.9	26.2	90.9	75.6	74.5	66.6
3	52.9	40.2	39.6	30.9	79.2	74.0	74.0	69.6
4	53.5	40.1	41.4	28.7	79.4	74.0	74.8	68.3
5	45.7	39.5	40.1	31.1	75.5	73.7	74.2	69.7
6	44.6	39.1	39.9	33.3	76.1	73.7	74.1	70.9



(a) Uniaxial compression (b) Conventional triaxial compression (60 MPa)

Fig. 10. Failure states of discontinuities in the equivalent fractured rock mass.
Note: "n" and "p" refer to "now" and "past", respectively. For example, "tension-n slip-p" indicates that the sub-contact has experienced slip failure in the past and is now in a state of tensile failure.



(a) Uniaxial compression (b) Conventional triaxial compression (60 MPa)

Fig. 11. Failure states of rock matrix zones in the equivalent fractured rock mass.
Note: Lode angle is calculated from the stress state of the rock matrix after yielding.

observed in Fig. 11, to reflect the specific stress state distribution ranging from tensile failure (0° to 30°) to shear failure (−30° to 0°).

Under uniaxial compression, as observed in Fig. 10a, three main through-going fractures are detected, with the corresponding failure state being primarily tensile failure. Meanwhile, Fig. 11a reveals that only a small portion of rock matrix yielded after rock mass failure, predominantly exhibiting shear failure. This is attributed to stress concentrations caused by the presence of discontinu-

ities. While under a confining pressure of 60 MPa, Fig. 10b demonstrates that the failure of discontinuities is primarily shear-driven, and no through-going fractures are observed. Furthermore, Fig. 11b shows that a significant portion of rock matrix underwent shear failure, with Lode angles mainly ranging between −30° and −20°. These results demonstrate that under uniaxial compression, fractured rock mass predominantly experiences tensile failure along discontinuities, where the strength of discontinuities controls failure behavior. In contrast, under high confining pressure, the failure mode shifts to shear failure, and the role of rock matrix becomes significant.

5.2. Impacts of the plastic flow rule of rock matrix

5.2.1. Strength and GSI of rock mass

When the non-associated flow coefficient β of rock matrix is set to 0, 0.2, 0.4, 0.6, 0.8, and 1, respectively, the strength and *GSI* of rock mass under uniaxial compression are plotted in Fig. 12, with the corresponding data provided in Table 5. It can be observed that the UCS and *GSI* decrease as the dilatancy degree of rock matrix increases, but the magnitude of reduction is relatively small. Specifically, when β increases from 0 (constant-volume flow rule) to 1 (associated flow rule), the predicted UCS decreases from 39.9 to 39.1 MPa, reduced by 2.0%. Correspondingly, *GSI* decreases from 74.1 to 73.8, with a reduction of only 0.5%. This phenomenon is attributed to the fact that, under unconfined compression conditions, fractured rock mass tends to undergo splitting failure along discontinuities. When associated law is applied to the rock matrix, the volumetric deformation is most severe. However, internal discontinuities, which have weak tensile resistance, struggle to bear significant lateral deformation. As a result, the discontinuities are more prone to tensile failure, leading to the loss of bearing capacity of rock mass.

When various confining pressures were applied, the strength and *GSI* of rock mass obtained when using constant-volume law and associated law for the rock matrix are shown in Fig. 13, with data provided in Table 6. It is crucial to note that the *GSI* values are equivalent (or apparent) *GSI* back-calculated from the simulated strength of the rock mass and the smooth GZZ criterion. While the intrinsic *GSI*, determined from geological mapping, remains constant. Comparative analysis reveals that, unlike the uniaxial compression scenario, the strength and *GSI* are significantly higher when associated flow rule is adopted for the rock matrix. The fundamental reason for this is the change in the failure mechanism of rock mass. Specifically, the confining pressure restricts its lateral deformation, making it less likely for internal discontinuities to undergo tensile failure. Under this condition, rock mass failure is primarily associated with the rock matrix. On the other hand, the rock matrix zones can experience mutual squeezing since associated flow rule induces greater volumetric deformation. This, to some extent, increases the effective confining pressure acting on the rock matrix zones, thereby enhancing their strength. Consequently, the calculated bearing capacity of rock mass is higher.

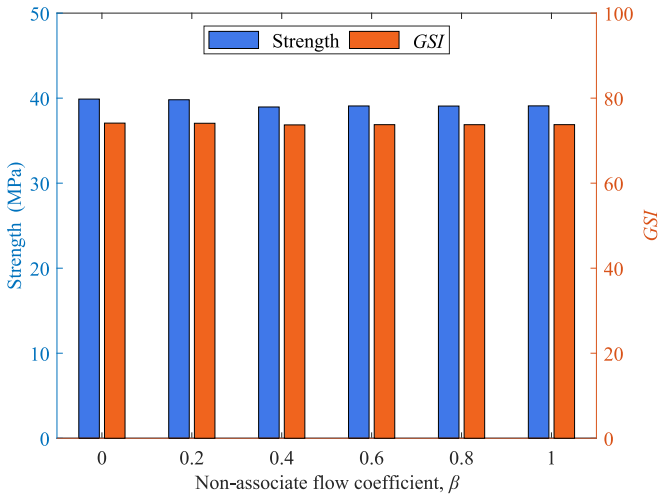


Fig. 12. Variation of equivalent parameters with non-associated flow coefficient.

Further analysis indicates that with increasing stress level, both the strength and *GSI* under the two rock matrix laws exhibit a nonlinear increase trend. Taking *GSI* as an example, while the confining pressure increases from 0 to 60 MPa, the *GSI* obtained using constant-volume flow rule for the rock matrix increases from 74.1 to 91.4, representing an increase of 23.3%. While the *GSI* obtained using associated flow rule increases from 73.8 to 97.1, with an increase of 31.6%, which is very close to 100 for intact rock. This phenomenon results from the gradual closure of discontinuities as the confining pressure increases, which progressively reduces their influences on the overall strength. Consequently, the strength of rock mass becomes increasingly closer to that of intact rock.

5.2.2. Deformation behavior of rock mass

Fig. 14a–b displays the stress–strain relations of the equivalent fractured rock mass when the rock matrix follows constant-volume law and associated law, respectively. It is obvious that the axial stress increases approximately linearly under both conditions before reaching its peak value. The minimum principal strain increases significantly near the peak strength, indicating that the equivalent fractured rock mass primarily undergoes splitting failure along the direction of minimum principal stress. This also explains why the calculated strength is lower when associated law is used for the rock matrix. Furthermore, even when constant-volume law is employed for the rock matrix, the rock mass still exhibits significant post-peak dilatancy behavior. This is because, under unconfined compression conditions, the primary reason for the loss of bearing capacity is the tensile failure of discontinuities, which leads to a sharp increase in lateral deformation, while the deformation behavior of rock matrix has a relatively minor influence.

Fig. 15 plots the axial stress–strain relations of the equivalent fractured rock mass calculated under various flow rules for the rock matrix. It can be observed that regardless of the flow rule adopted, the failure mode of rock mass under uniaxial compression is brittle. In contrast, under high confining pressure, the axial stress–strain curves of the equivalent fractured rock mass approach elastic perfectly-plastic, with the corresponding failure mode being ductile. This reveals the influence of confining pressure on the brittle–ductile transition of rock mass. Specifically, at low confining pressure, brittle failure usually takes place, while as the stress level increases, brittle failure gradually transitions to ductile failure.

5.3. Impacts of the strength criterion of rock matrix

5.3.1. Strength and GSI of rock mass

Adopting the elastoplastic constitutive model based on the H–B criterion for the rock matrix while using constant-volume flow rule, uniaxial and conventional triaxial compression experiments were conducted successively. The obtained strength and *GSI* of rock mass are displayed in Fig. 16, with specific data provided in Table 6.

From the uniaxial compression results, the strength obtained using the H–B strength criterion for the rock matrix (47.4 MPa)

Table 5
UCS and *GSI* of the equivalent fractured rock mass with various non-associated flow coefficients β.

β	0	0.2	0.4	0.6	0.8	1.0
Strength (MPa)	39.9	39.8	39.0	39.1	39.1	39.1
GSI	74.1	74.1	73.7	73.8	73.8	73.8

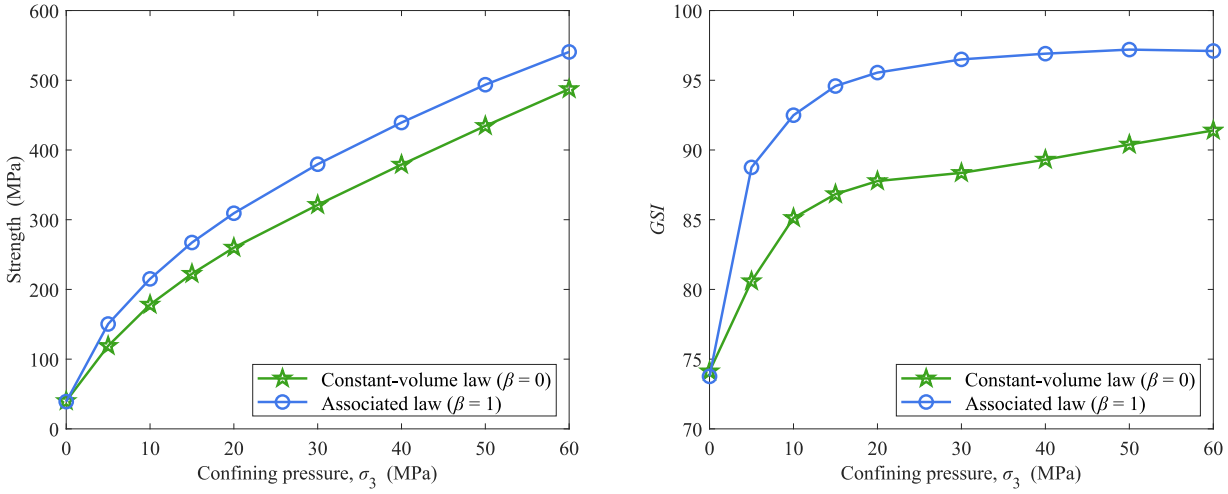


Fig. 13. Variation of equivalent parameters with confining pressure under different flow rules.

Table 6
UCS and *GSI* of the equivalent fractured rock mass with various confining pressures.

Confining pressure (MPa)	Constant-volume law (GZZ)		Associated law (GZZ)		Constant-volume law (H–B)	
	Strength (MPa)	*GSI*	Strength (MPa)	*GSI*	Strength (MPa)	*GSI*
0	39.9	74.1	39.1	73.8	47.4	77.2
5	119.0	80.6	150.4	88.8	97.8	72.5
10	178.2	85.1	215.1	92.5	137.3	73.0
15	222.5	86.8	267.3	94.6	170.7	73.6
20	260.0	87.8	309.3	95.6	200.5	74.2
30	321.3	88.4	379.6	96.5	259.4	76.5
40	379.0	89.3	439.3	96.9	316.2	79.0
50	434.5	90.4	493.6	97.2	365.0	80.2
60	487.4	91.4	540.6	97.1	414.8	81.8

is 18.8% higher than that obtained using the GZZ strength criterion (39.9 MPa), and the corresponding *GSI* is 4.1% higher. This anomalous outcome occurs because, under uniaxial compression conditions, the failure is less influenced by the rock strength and is primarily caused by tensile failure of discontinuities, as revealed in Fig. 10a and Fig. 11a. Further comparison shows that although both strength criteria adopt constant-volume flow rule, the difference lies in the distribution ratio of lateral deformation after yielding of the rock matrix. As mentioned earlier, greater lateral deformation makes the discontinuities perpendicular to that direction more prone to tensile failure. Therefore, in the direction perpendicular to the fracture plane, if the deformation calculated by the GZZ model happens to be larger than that by the H–B model, it will result in a lower predicted strength when the GZZ strength criterion is used for the rock matrix. In fact, in the simulations on other equivalent fractured rock mass models, cases were also observed where the use of the GZZ strength criterion for the rock matrix led to higher predicted UCS and *GSI* values.

When the confining pressure varies from 0 to 60 MPa, similar to the case when the GZZ strength criterion is adopted, the predicted strength of the equivalent fractured rock mass using the H–B strength criterion also increases nonlinearly. However, the strength values are significantly lower than those predicted by the GZZ criterion. For example, at a confining pressure of 60 MPa, the strength obtained using the H–B criterion (414.8 MPa) is 14.9% lower than that predicted by the GZZ criterion (487.4 MPa). Regarding the *GSI* values back-calculated from the strength, unlike the case with the GZZ criterion, the predictions hereby first decrease and then increase, making it hard to explain the continuous hardening of rock mass strength parameters with increasing confining pressure.

5.3.2. Deformation performance of rock mass

In terms of deformation, Fig. 17a presents the stress–strain relation of the equivalent fractured rock mass under uniaxial compression when the H–B strength criterion is adopted. Comparing Fig. 14a with Fig. 17a, it can be found that the deformation curves based on the H–B criterion are similar to those obtained with the GZZ criterion. Both exhibit an approximately linear increase in axial stress with strain until brittle failure occurs near the peak strength. Fig. 17b plots the axial stress–strain relations of the equivalent fractured rock mass under uniaxial and conventional triaxial compression when the H–B strength criterion is applied. Compared with the results obtained using the GZZ strength crite-

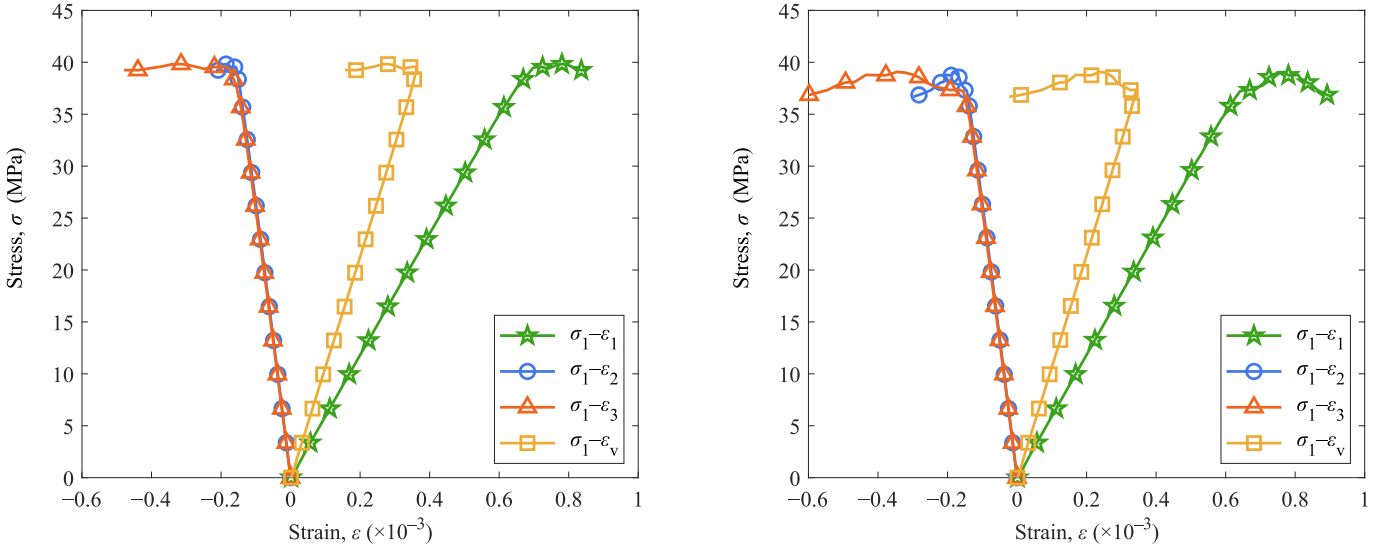


Fig. 14. Stress–strain curves of the equivalent fractured rock mass under uniaxial compression.

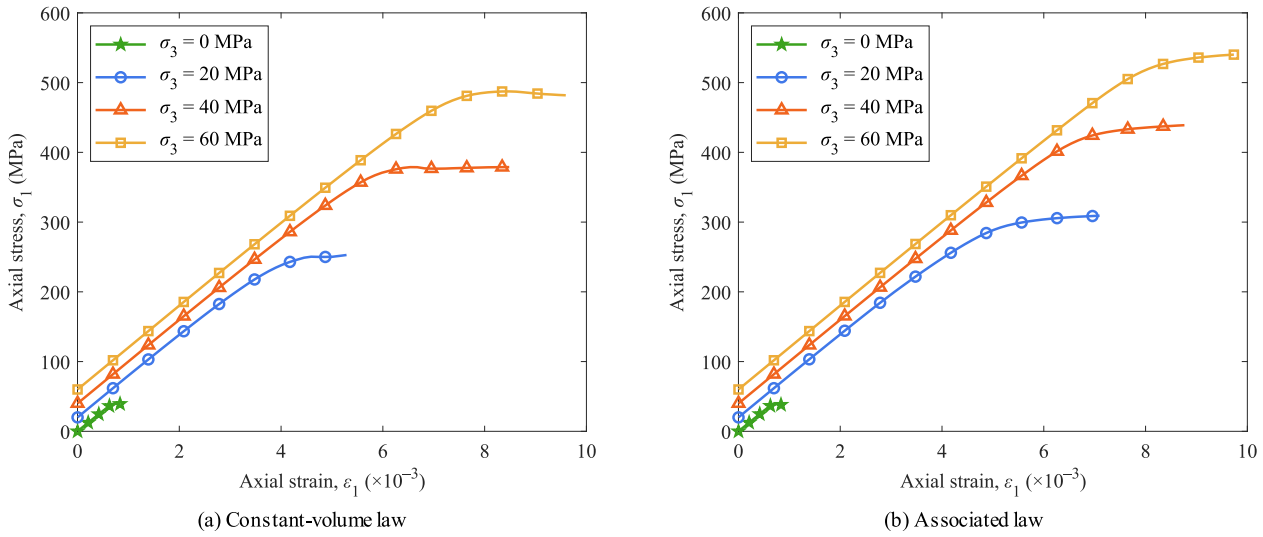


Fig. 15. Axial stress–strain curves of the equivalent fractured rock mass under compression.

rion, as indicated in Fig. 15a, it is found that when the stress level increases, the axial stress no longer undergoes a sharp drop after reaching its peak value but gradually approaches elastic perfect-plastic behavior. This is due to the increasing contribution of rock matrix, which causes the failure mode of the equivalent fractured rock mass to transition gradually from brittle to ductile.

6. Conclusions

(1) This study proposes an elastoplastic constitutive model based on the smooth GZZ strength criterion for rock materials., which was validated by comparing the prediction of strength with the uniaxial and true triaxial compression experimental results of the intact rock and two types of prefabricated fractured rock specimens. The predicted strength of the prefabricated fractured rock specimens shows good agreement with experimental results, indicating that the model is capable of estimating the strength and *GSI* of fractured rock mass. Case analysis as well as numerical results demonstrates that both the UCS and *GSI* of the rock mass ahead of a deep tunnel face exhibit significant size effect. Specifically, the average *GSI* value obtained from numerical tests on the equivalent fractured rock mass with a dimension of 6 m was 73.7, and the average UCS decreased by more than 70% compared to that of intact rock.

(2) Under unconfined compression, fractured rock mass primarily undergoes tensile failure along discontinuities, with only localized shear yielding in a small portion of rock matrix. In such case, the strength of discontinuities dominates, and the failure mode of rock mass is brittle. As the confining pressure increases, tensile failure no longer occurs along the discontinuities, while the proportion of shear failure in the rock matrix zones gradually rises. The axial stress–strain relation of the equivalent fractured rock mass progressively

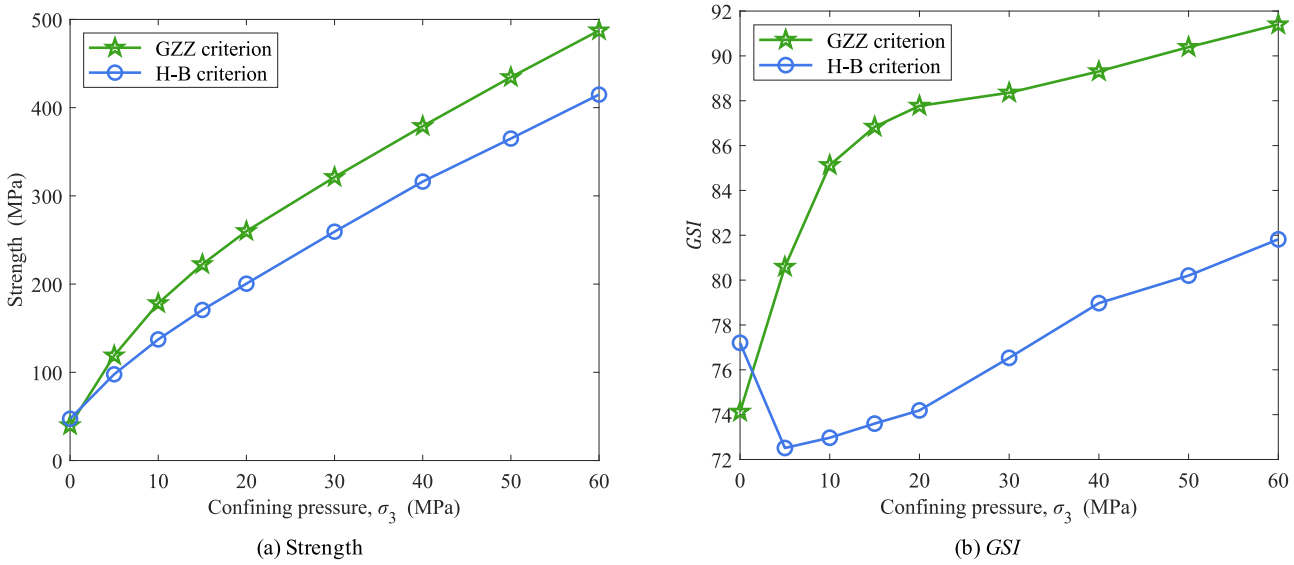


Fig. 16. Variation of equivalent parameters with confining pressure under different criteria.

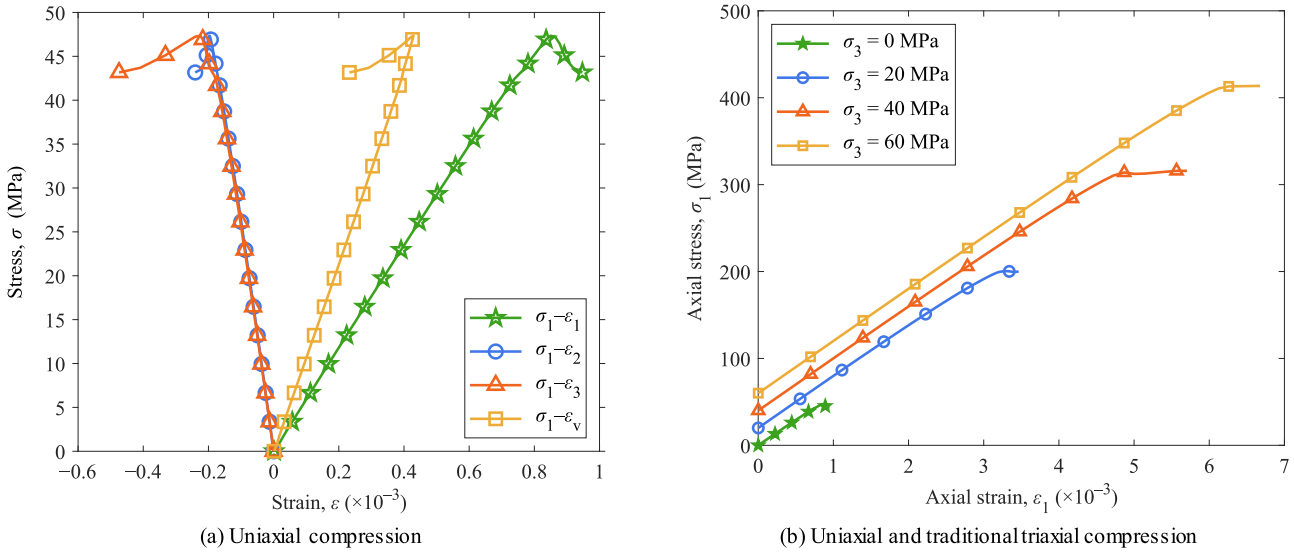


Fig. 17. Stress–strain curves of the equivalent fractured rock mass when the H–B criterion is adopted.

approaches elastic perfect-plastic behavior, and the failure mode transitions from brittle to ductile. This reflects the significant influence of confining pressure on the brittle–ductile transition in fractured rock mass.

(3) Under unconfined compression, the strength and *GSI* of the equivalent fractured rock mass calculated using constant-volume flow rule (which neglects the dilatancy effects) for the rock matrix are slightly higher than those obtained using associated flow rule (which maximizes the dilatancy effects), but the difference is negligible. As confining pressure increases, both the strength and *GSI* increase nonlinearly. The rise in *GSI* reflects the continuous enhancement of rock mass strength parameters under confining pressure, indicating that the traditional *GSI* determination method requires improvement under high-stress conditions. However, when confining pressure is applied, the strength and *GSI* calculated using associated flow rule are significantly higher than those obtained with constant-volume flow rule. This suggests that considering the dilatancy effects of rock matrix under high confining pressure leads to overestimated strength predictions.

(4) The selection of strength criterion for the rock matrix significantly influences the predicted results of strength and *GSI* of rock mass. Both the GZZ and H–B strength criteria can capture the brittle-ductile transition in fractured rock mass as well as the nonlinear enhancement in strength with rising stress level. However, the H–B criterion fails to describe the continuous hardening process of rock mass strength parameters. Moreover, since the H–B criterion neglects the beneficial effect of σ_2, the predicted strength of the equivalent fractured rock mass is conservative under high stress level. For example, in the studied case, at a confining pressure of 60 MPa, the predicted strength using the H–B criterion was 14.9% lower than that obtained with the GZZ criterion.

However, this study has certain limitations: (1) The elastoplastic model for the intact rock may not fully capture the post-peak strain-softening behavior of rock mass. (2) The predicted equivalent mechanical parameters of the rock mass have not yet been directly validated by large-scale in-situ or laboratory tests. (3) The model does not consider the influences of groundwater, time-dependent deformation, or other environmental coupling effects.

Future work should focus on the following directions to advance this research: (1) Conducting large-scale true triaxial experiments on fractured rock mass to provide direct validation for the proposed model under complex stress paths. (2) Investigating specifically the failure mechanisms of deep rock mass under excavation-induced unloading, thereby providing direct guidance for support design optimization. (3) Developing a robust implicit stress integration algorithm based on advanced 3-D strength criteria to significantly enhance computational efficiency for models containing numerous discontinuities. This development is a crucial step towards establishing a rapid analysis framework—from excavation face data acquisition to rock mass parameter estimation—thereby paving the way for adaptive design and enhanced hazard control in deep underground engineering.

CRediT authorship contribution statement

Xiangyang Wei: Writing – original draft, Visualization, Validation, Methodology, Investigation. **Wuqiang Cai:** Methodology, Funding acquisition, Conceptualization. **Hehua Zhu:** Supervision, Project administration. **Wenhao Liang:** Supervision, Conceptualization. **Xiaojun Wang:** Formal analysis, Data curation. **Jinfeng Xu:** Investigation, Data curation. **Fengshou Zhang:** Writing – review & editing, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by National Natural Science Foundation of China (No. 42307206).

References

[1] Cai WQ, Zhu HH, Liang WH, Vu B, Su CL, Zhang KS, et al. Three-dimensional forward analysis and real-time design of deep tunneling based on digital *in-situ* testing. Int J Mech Sci 2022;226:107385.
[2] Guo SF, Qi SW, Zhan ZF, Ma LN, Gure EG, Zhang SS. Numerical study on the progressive failure of heterogeneous geomaterials under varied confining stresses. Eng Geol 2020;269:105556.
[3] Zhu HH, Yan JX, Liang WH. Challenges and development prospects of ultra-long and ultra-deep mountain tunnels. Engineering 2019;5(3):384–92.
[4] Feng CC, Wang ZL, Wang JG, Lu ZT, Li SY. A thermo-mechanical damage constitutive model for deep rock considering brittleness-ductility transition characteristics. J Cent South Univ 2024;31(7):2379–92.
[5] Gao YH, Wang KZ, Zhou C. A numerical study on true triaxial strength and failure characteristics of jointed marble. Acta Geotech 2022;17(5):2001–20.
[6] Gao XS, Wang M, Li C, Zhang MM, Li ZH. A new three-dimensional rock strength criterion based on shape function in deviatoric plane. Geomech Geophys Geo Energy Geo Resour 2024;10(1):7.
[7] Hoek E, Brown ET. Empirical strength criterion for rock masses. J Geotech Engrg Div 1980;106(9):1013–35.
[8] Singh A, Ayothiraman R, Rao KS. Failure criteria for isotropic rocks using a smooth approximation of modified Mohr–coulomb failure function. Geotech Geol Eng 2020;38(4):4385–404.
[9] Yu MH. Unified strength theory and its applications. Singapore: Springer Singapore; 2018.
[10] Liu HT, Han Z, Han ZJ, Chen ZH, Liu QY, Zhang HK, et al. Nonlinear empirical failure criterion for rocks under triaxial compression. Int J Min Sci Technol 2024;34:351–69.
[11] Hoek E. Hoek-Brown failure criterion-2002 edition. Proc Fifth North Am Rock Mech Symp 2002;1:18–22.
[12] Hoek E, Brown ET. The Hoek–Brown failure criterion and GSI–2018 edition. J Rock Mech Geotech Eng 2019;11(3):445–63.
[13] Hoek E, Marinos PG. Predicting tunnel squeezing problems in weak heterogeneous rock masses. Tunn Tunn Int 2000;32:45–51.
[14] Cai WQ, Zhu HH, Liang WH, Wang XJ, Su CL, Wei XY. A post-peak dilatancy model for soft rock and its application in deep tunnel excavation. J Rock Mech Geotech Eng 2023;15(3):683–701.
[15] Cai WQ, Zhu HH, Liang WH. Three-dimensional tunnel face extrusion and reinforcement effects of underground excavations in deep rock masses. Int J Rock Mech Min Sci 2022;150:104999.
[16] Zhang LY, Zhu HH. Three-dimensional hoek-brown strength criterion for rocks. J Geotech Geoenviron Eng 2007;133(9):1128–35.
[17] Wang ZL, Wang HC, Wang JG, Tian NC. Finite element analyses of constitutive models performance in the simulation of blast-induced rock cracks. Comput Geotech 2021;135:104172.
[18] Zheng Z, Xu HY, Zhang K, Feng GL, Zhang Q, Zhao YF. Intermittent disturbance mechanical behavior and fractional deterioration mechanical model of rock under complex true triaxial stress paths. Int J Min Sci Technol 2024;34:117–36.
[19] Aladejare AE, Wang Y. Influence of rock property correlation on reliability analysis of rock slope stability: from property characterization to reliability analysis. Geosci Front 2018;9(6):1639–48.
[20] Zheng MZ, Li SJ, Zhao HB, Huang X, Qiu SL. Probabilistic analysis of tunnel displacements based on correlative recognition of rock mass parameters. Geosci Front 2021;12(4):101136.
[21] Jia CJ, Chen FL, Zhou SJ, Lei MF, Huang J, Zheng YN. Effect of joint density on mechanical behavior of rock mass: Insight from 3D printing tests and DEM simulation. Eng Fract Mech 2025;315:110846.
[22] Wu AQ, Fan L, Fu X, Zhang YH, Zhong ZW, Yu MW. Design and application of hydro-mechanical coupling test system for simulating rock masses in high dam reservoir operations. Int J Rock Mech Min Sci 2021;140:104638.
[23] Liu ZJ, Zhang CS, Zhang CQ, Wang HB, Zhou H, Zhou B. Effects of amygdale heterogeneity and sample size on the mechanical properties of basalt. J Rock Mech Geotech Eng 2022;14(1):93–107.
[24] Farahmand K, Vazaios I, Diederichs MS, Vlachopoulos N. Investigating the scale-dependency of the geometrical and mechanical properties of a moderately jointed rock using a synthetic rock mass (SRM) approach. Comput Geotech 2018;95:162–79.
[25] Zhou YJ, Feng WK, Hu YP, Yi XY, Ji F, Li WB. Equivalent continuous numerical simulation of a large-scale underground powerhouse excavation considering the size effect of the jointed rock mass. Tunn Undergr Space Technol 2024;154:106058.
[26] Zhou ZQ, Sun JW, Lai YB, Wei CC, Hou J, Bai SS, et al. Study on size effect of jointed rock mass and influencing factors of the REV size based on the SRM method. Tunn Undergr Space Technol 2022;127:104613.
[27] Wu N, Liang ZZ, Zhang ZH, Li SH, Lang YX. Development and verification of three-dimensional equivalent discrete fracture network modelling based on the finite element method. Eng Geol 2022;306:106759.
[28] Zhang Q, Wang XJ, Zhu HH, Zhang KS, Li XJ. Mixture distribution model for three-dimensional geometric attributes of multiple discontinuity sets based on trace data of rock mass. Eng Geol 2022;311:106915.
[29] Zhang Q, Wang XJ, Zhu HH, Cai WQ, Li XJ. Estimation of three-dimensional diameter distributions of multiple fracture sets clustered by a multi-level clustering method. Acta Geotech 2023;18(8):4429–52.
[30] Zhang L. A generalized three-dimensional Hoek–Brown strength criterion. Rock Mech Rock Eng 2008;41(6):893–915.
[31] Cai WQ, Zhu HH, Liang WH, Zhang LY, Wu W. A new version of the generalized Zhang–Zhu strength criterion and a discussion on its smoothness and convexity. Rock Mech Rock Eng 2021;54(8):4265–81.
[32] Vermeer PA. Non-associated plasticity for soils, concrete and rock. Physics of Dry Granular Media. Dordrecht: Springer Netherlands; 1998:163–96.
[33] Zhu HH, Zhang Q, Huang BQ, Zhang LY. A constitutive model based on the modified generalized three-dimensional Hoek–Brown strength criterion. Int J Rock Mech Min Sci 2017;98:78–87.
[34] Itasca Consulting Group I. 3DEC — Three-Dimensional Distinct Element Code, Version 7.0. Minneapolis, MN, USA: Itasca Consulting Group, Inc; 2020.
[35] Zhang KS, Wu W, Zhang M, Liu YS, Huang Y, Chen BL. New method to identify optimal discontinuity set number of rock tunnel excavation face orientation based on Fisher mixed evaluation. Undergr Space 2024;17:300–19.
[36] Barton N, Bandis S, Bakhtar K. Strength, deformation and conductivity coupling of rock joints. Int J Rock Mech Min Sci Geomech Abstr 1985;22(3):121–40.
[37] Park JW, Lee YK, Song JJ, Choi BH. A constitutive model for shear behavior of rock joints based on three-dimensional quantification of joint roughness. Rock Mech Rock Eng 2013;46(6):1513–37.
[38] Barton N, Choubey V. The shear strength of rock joints in theory and practice. Rock Mech 1977;10(1):1–54.